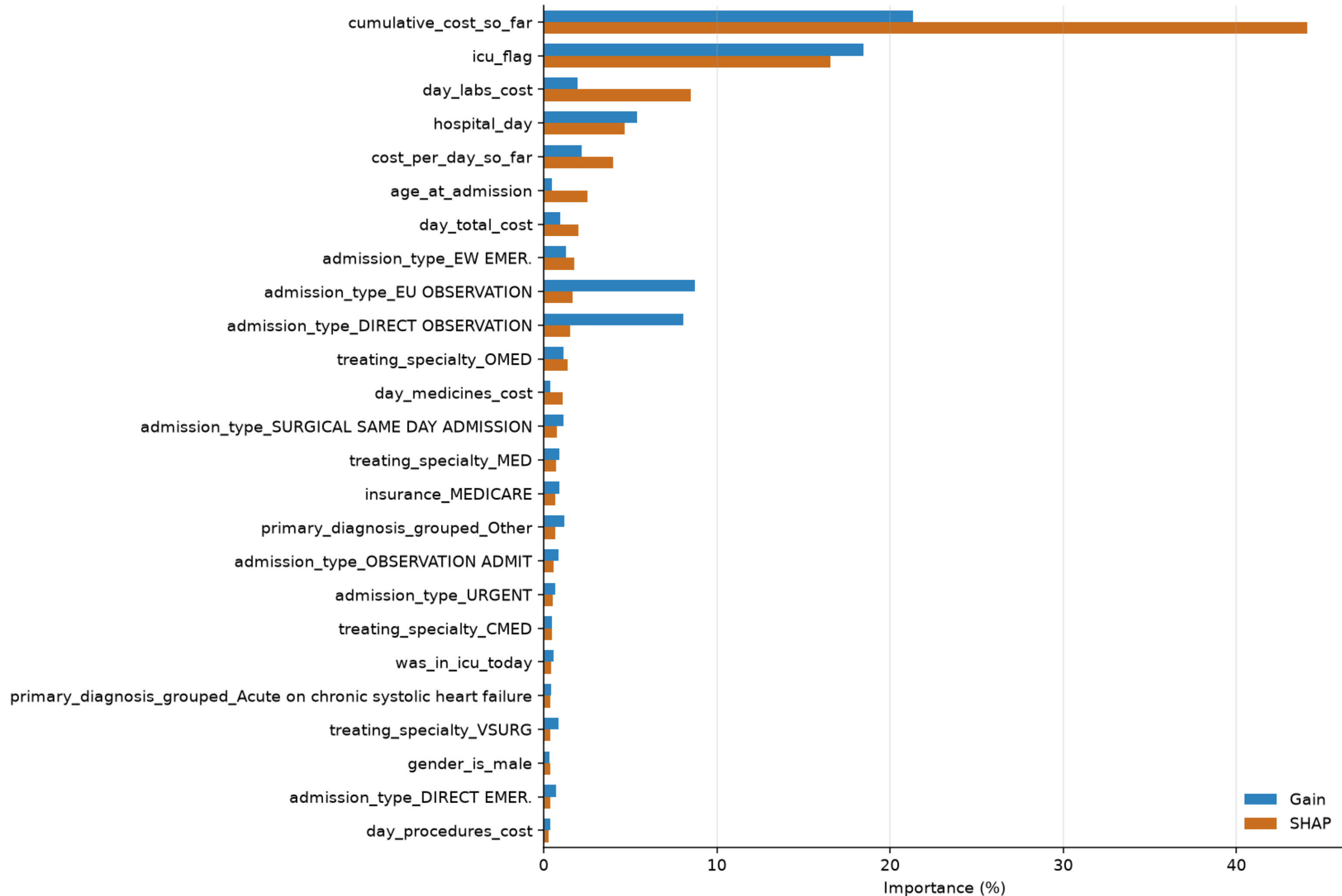
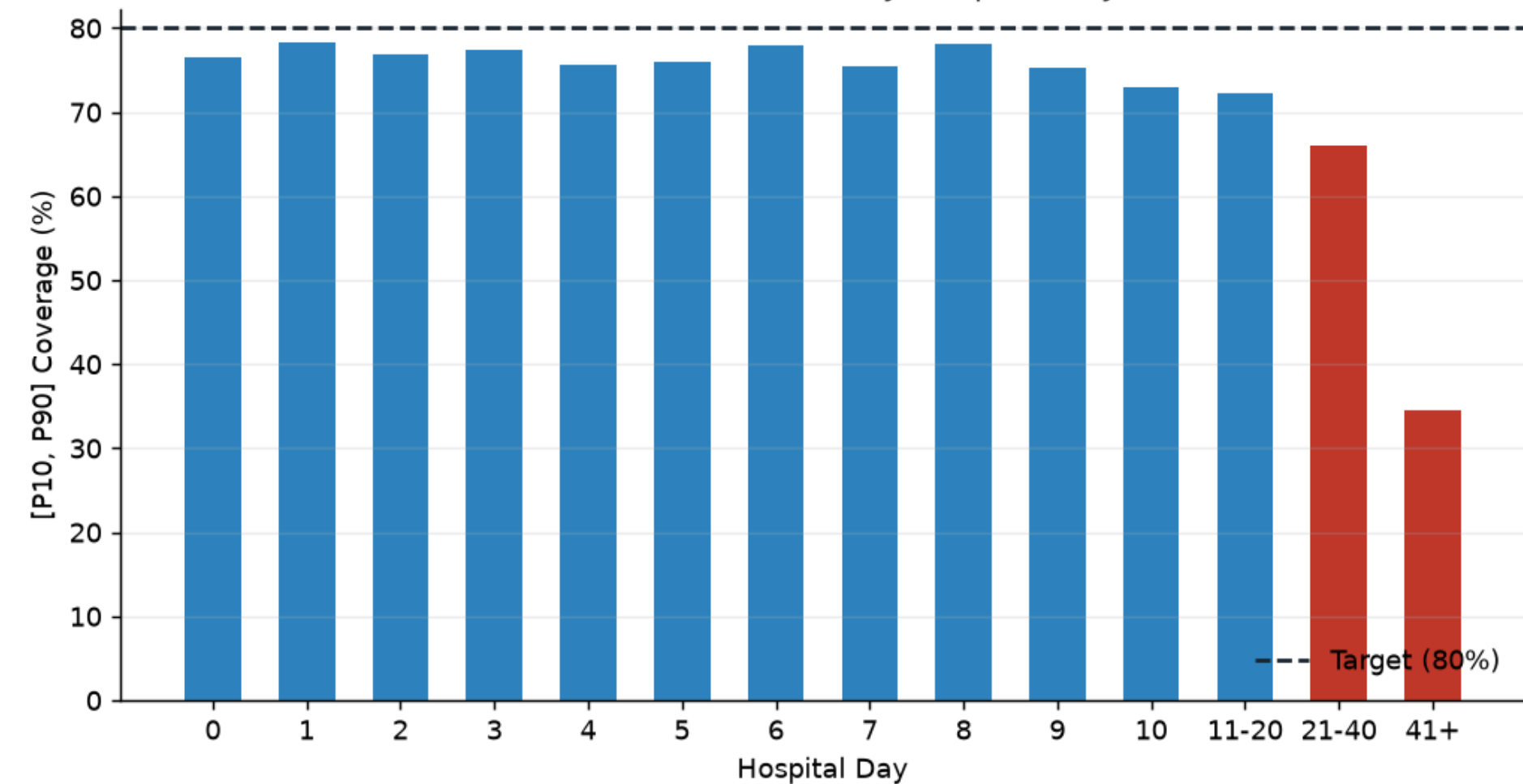


Quantile Model Eval

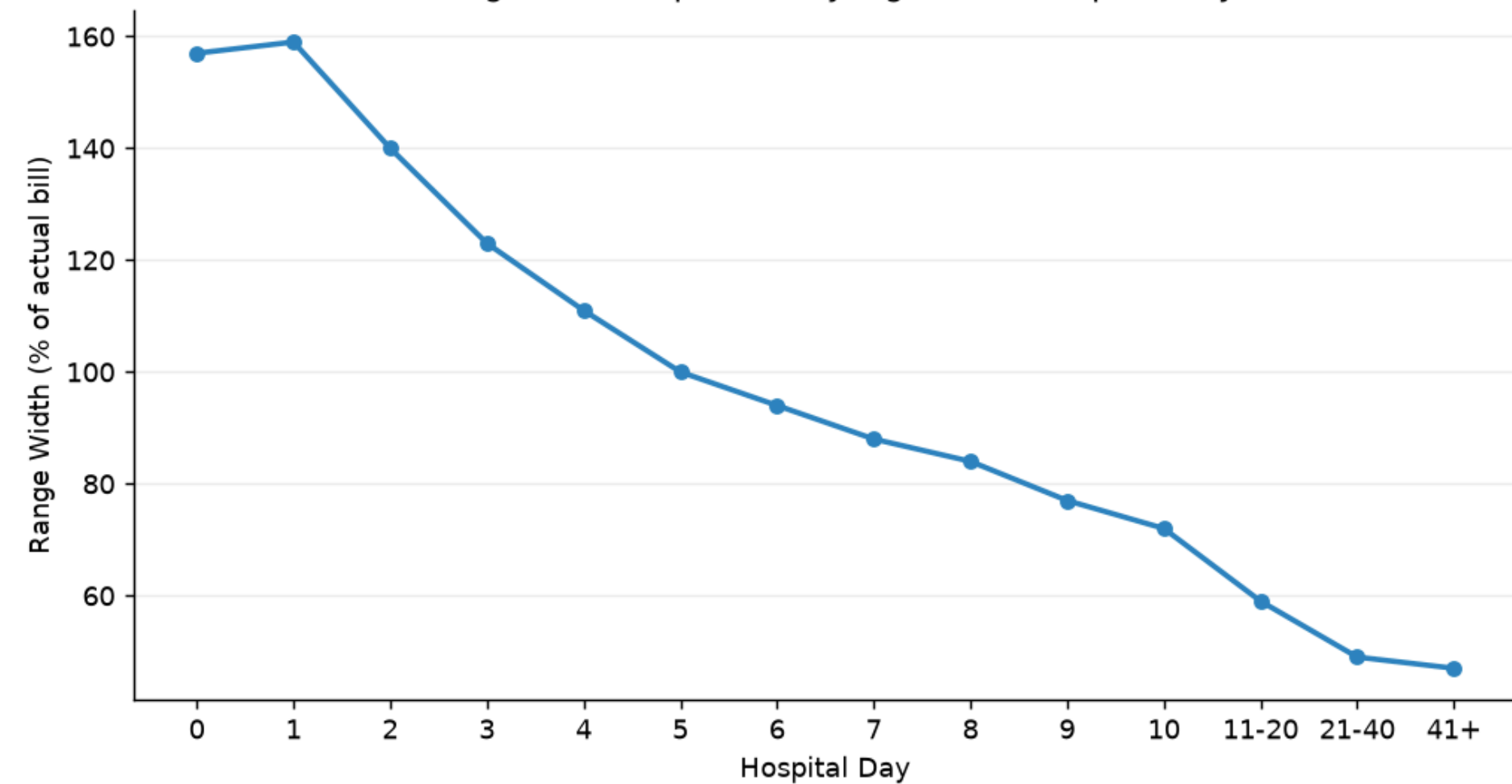
Top 25 Features (incl. day_procedures_cost, day_medicines_cost, gender_is_male) — Gain vs SHAP



Quantile Calibration by Hospital Day



Prediction Range Gets Proportionally Tighter as Hospital Day Increases

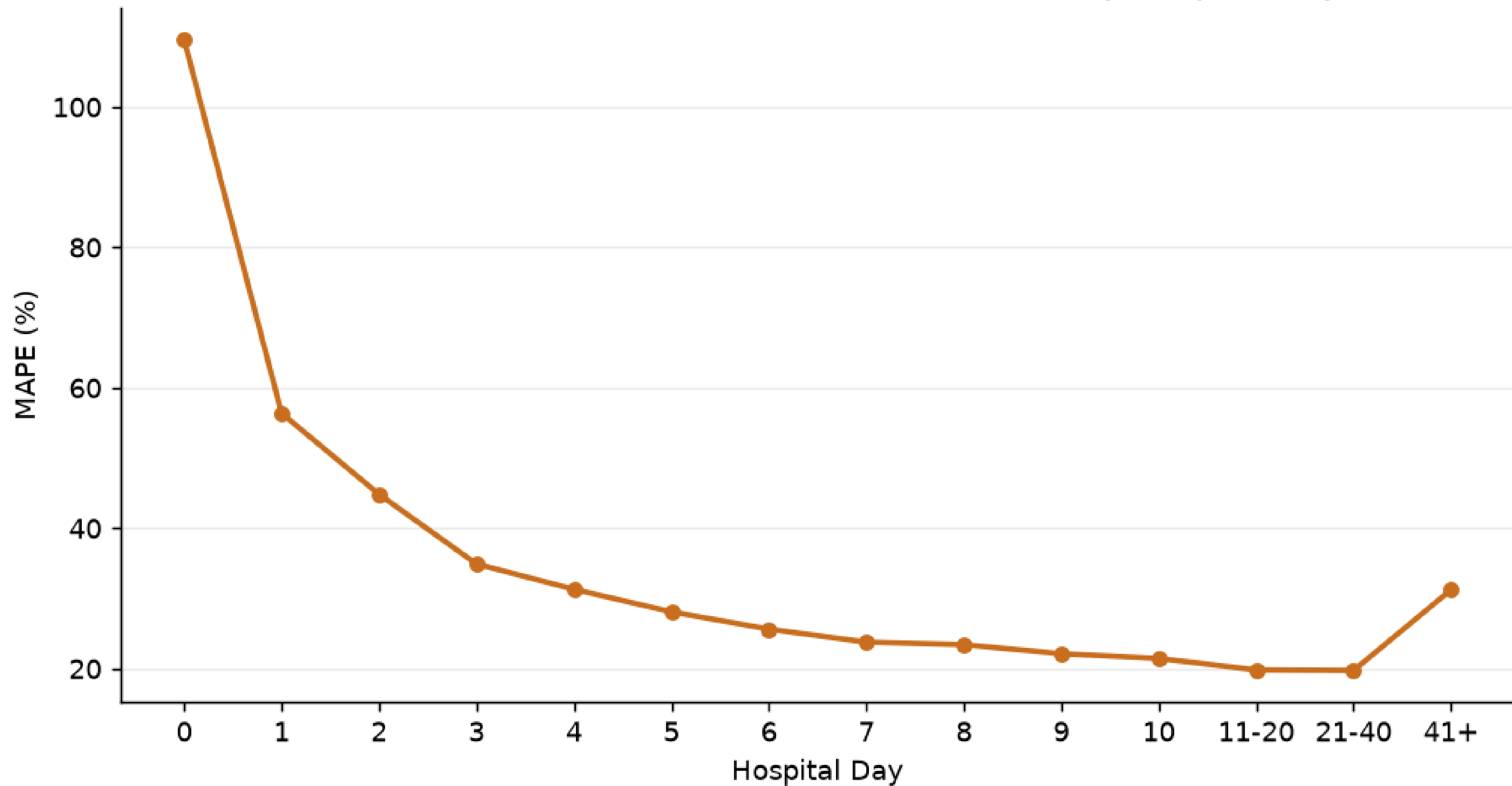


How Reliable Is Our Predicted Cost Range? Accurate Through Most of the Stay, Breaks Down Beyond 40 Days

day_bin	n	mean_true	mean_p10	mean_p50	mean_p90	coverage_80	interval_width	relative_width_pct	ceiling_floor_ratio
0	1469	176144.1	49549.9	113479.6	326161.6	0.764466	276611.7	157	6.58
1	1302	195275.8	74438.6	142589.5	385712.6	0.782642	311274	159	5.18
2	1070	228065.5	99458.21	168452.1	418634.4	0.769159	319176.2	140	4.21
3	869	265732.7	129323.4	194396.4	455971.4	0.774453	326648	123	3.53
4	715	305321.2	162951.1	229718.6	501762	0.756643	338810.8	111	3.08
5	587	345633.9	193261.7	264134.5	537975.2	0.759796	344713.5	100	2.78
6	476	392557.9	228031	304925.3	596253.1	0.779412	368222.1	94	2.61
7	388	441884.8	263825.9	347781.5	651159.9	0.755155	387334	88	2.47
8	330	483554	294737.7	388355.6	701062.6	0.781818	406324.9	84	2.38
9	280	527692.1	327590.6	428284.6	736192.6	0.753571	408602	77	2.25
10	241	574091.5	362297.4	460827.9	773253.3	0.73029	410955.8	72	2.13
11-20	1239	768388.3	484924.4	635875.7	940710.9	0.722357	455786.4	59	1.94
21-40	552	1349638	723477.3	1119607	1386748	0.65942	663270.9	49	1.92
41+	331	1801766	554913.4	1272886	1398667	0.344411	843753.8	47	2.52

Column	Meaning
day_bin	Stage of hospital stay (Day 0, 1, 2... or a range like "11-20")
n	Number of patient-days in this bucket
mean_true	Actual average final bill (₹)
mean_p10	Predicted low-end estimate (10% chance real bill is below this)
mean_p50	Predicted best-guess / median estimate
mean_p90	Predicted high-end estimate (10% chance real bill is above this)
coverage_80	% of real bills that actually fell inside [P10, P90] – target ~80%
interval_width	Range size in ₹ (mean_p90 – mean_p10)
relative_width_pct	Range size as % of the actual bill – shows true precision

Quantile Model (P50 / Median) — Relative Error by Hospital Day

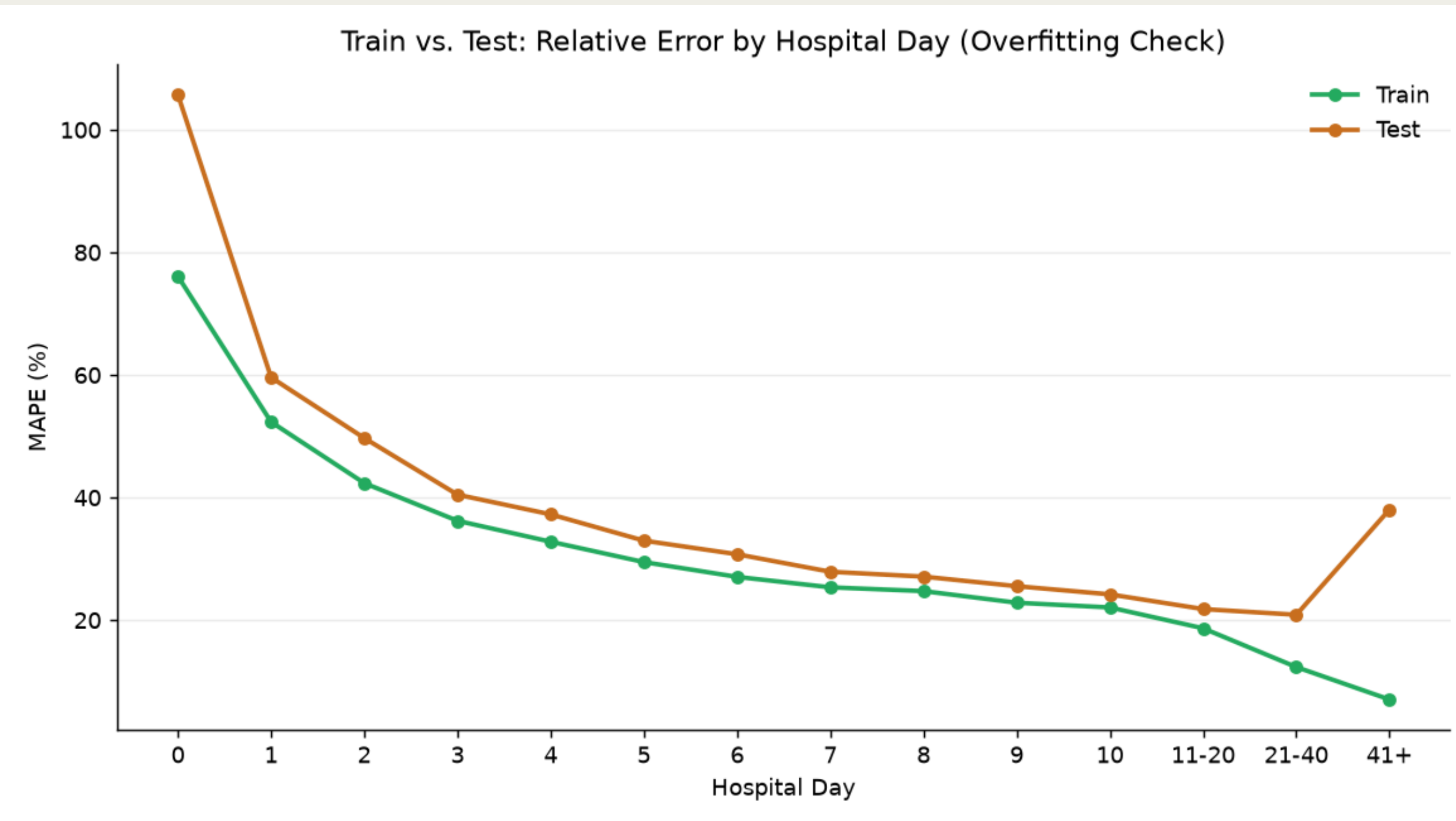


As more of the stay is observed, the model's prediction gets meaningfully more accurate relative to the bill size — up until you reach the rarest, longest stays, where accuracy dips again due to lack of training examples

day_bin	n	mae	mape	mean_true
0	1469	103132.1	105.7076	176144.1
1	1302	102442.1	59.58679	195275.8
2	1070	109979.6	49.69172	228065.5
3	869	115032.9	40.46223	265732.7
4	715	122765.6	37.248	305321.2
5	587	127135.5	32.9632	345633.9
6	476	133400.9	30.74324	392557.9
7	388	143107.3	27.88867	441884.8
8	330	151573.3	27.09611	483554
9	280	159631.1	25.53409	527692.1
10	241	167017	24.20202	574091.5
11-20	1239	201080.8	21.80191	768388.3
21-40	552	339372	20.87351	1349638
41+	331	629145.1	38.048	1801766

Column	Meaning
day_bin	Stage of hospital stay (Day 0, 1, 2... or a range like "11-20")
n	Number of patient-days in this bucket
mae	Mean Absolute Error – average ₹ the quantile model's median (P50) prediction was off by
mape	Mean Absolute Percentage Error – same error, as a % of the actual bill (comparable across days)
mean_true	Actual average final bill (₹) for patients in this bucket

Overfitting Check (Train V/S Test)



What we found:

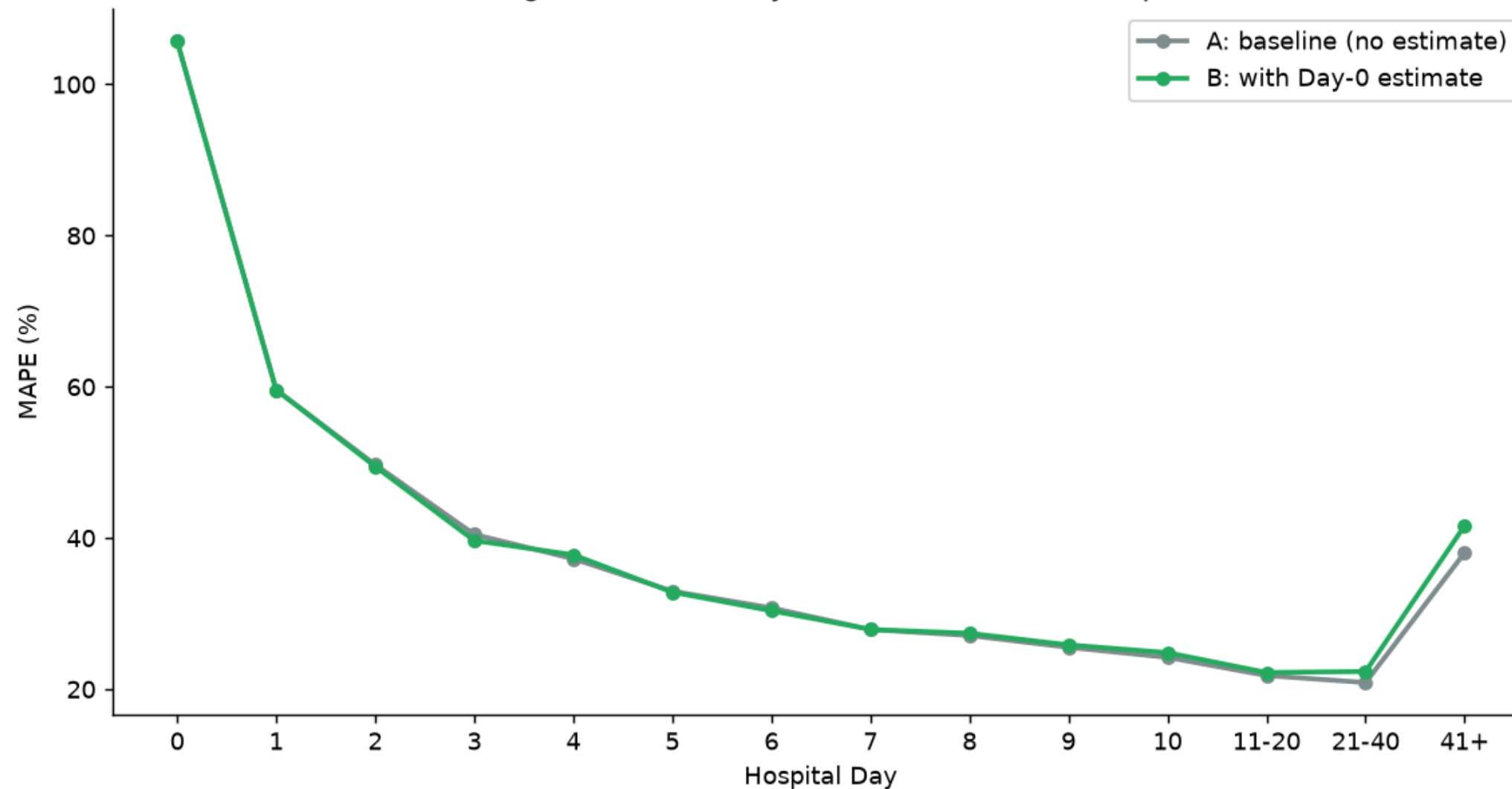
**Train MAPE = 38.4% overall,
Test MAPE = 47.3% overall**

**Test error is consistently higher
than train error at every single
day**

Experiment : 1

Does Our Own Day-0 Guess Help Later Predictions?

Does Feeding the Model's Day-0 Estimate Forward Help? (A vs B)



What was tested

- **A (baseline): predicts cost using only real clinical/demographic features — no memory of any earlier guess**
- **B (+ Day-0 estimate): same model, but also given the model's own end-of-Day-0 prediction as an extra input for every later day**

Result

- **Overall MAPE — A: 47.3% | B: 47.5%**
- **The two lines sit almost exactly on top of each other across every hospital day**

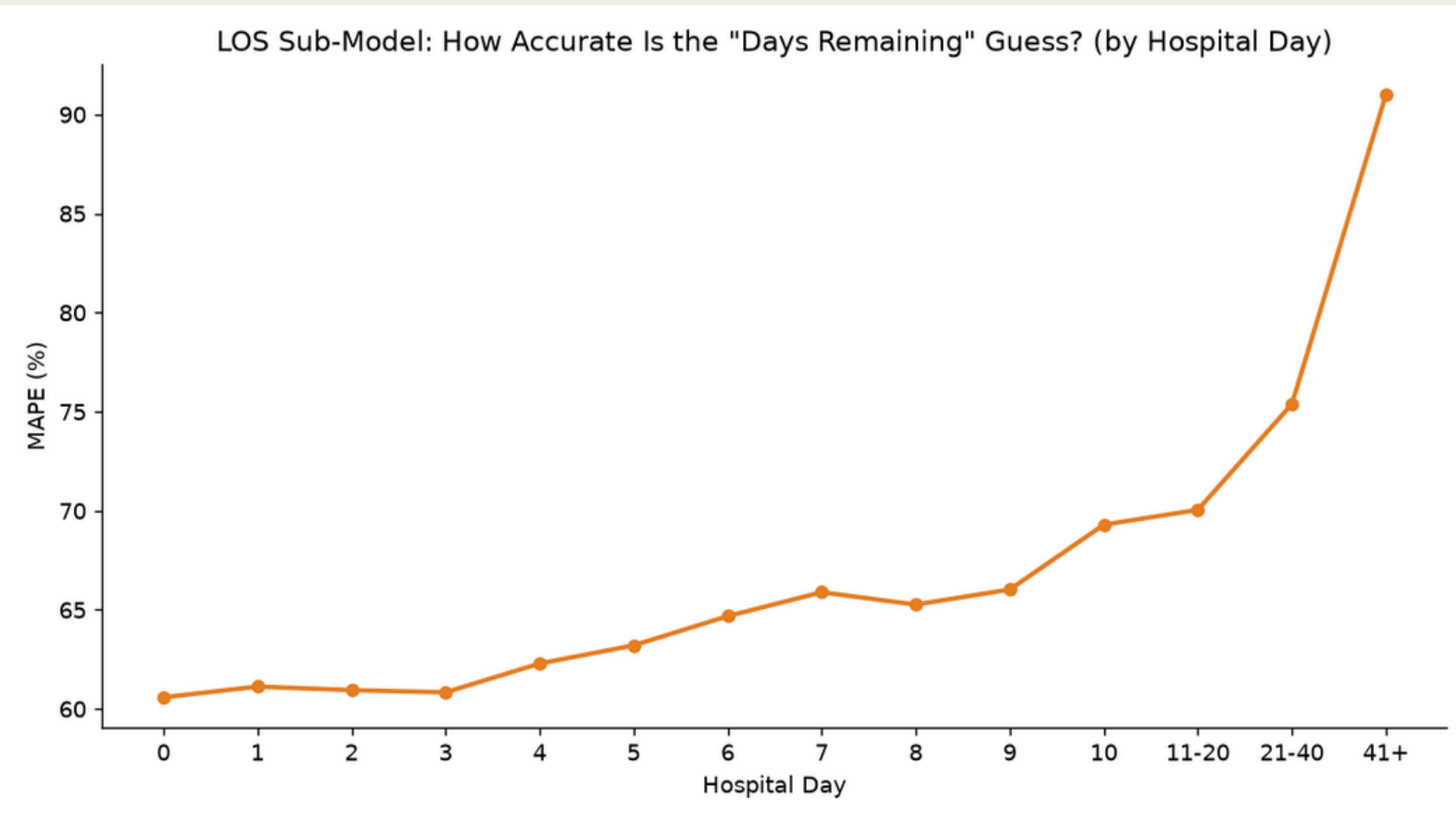
About the LOS Sub-Model – What Is It?

What it predicts:-

remaining_days = how many more days a patient will stay, predicted fresh from any point during their stay

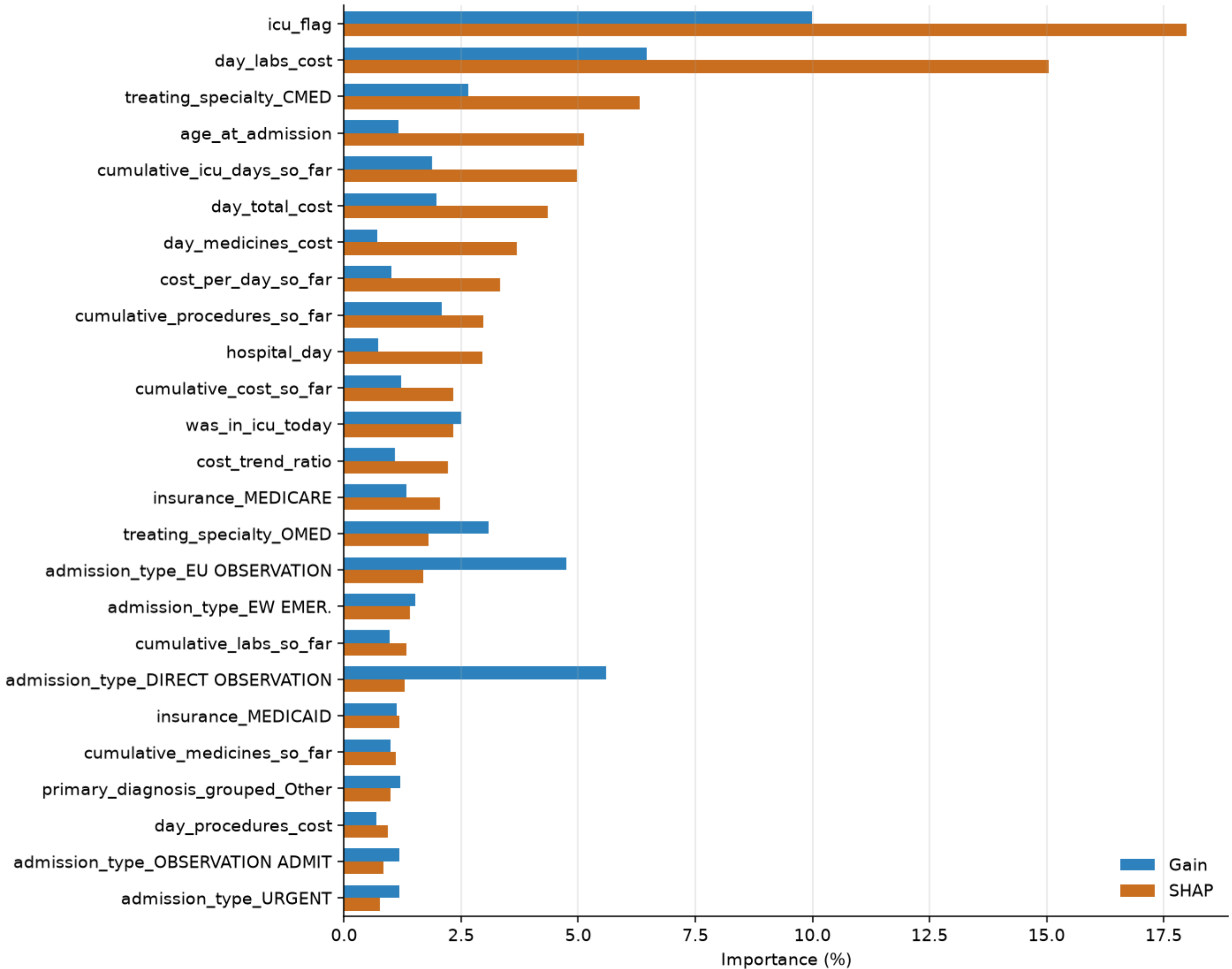
***MAPE (Mean Absolute Percentage Error)** = how far off the "days remaining" prediction is, as a % of the actual days remaining.*

***Formula:** $\text{actual_remaining_days} - \text{predicted_remaining_days} / \text{actual_remaining_days} \times 100$, averaged across all patients in that day-bucket.*

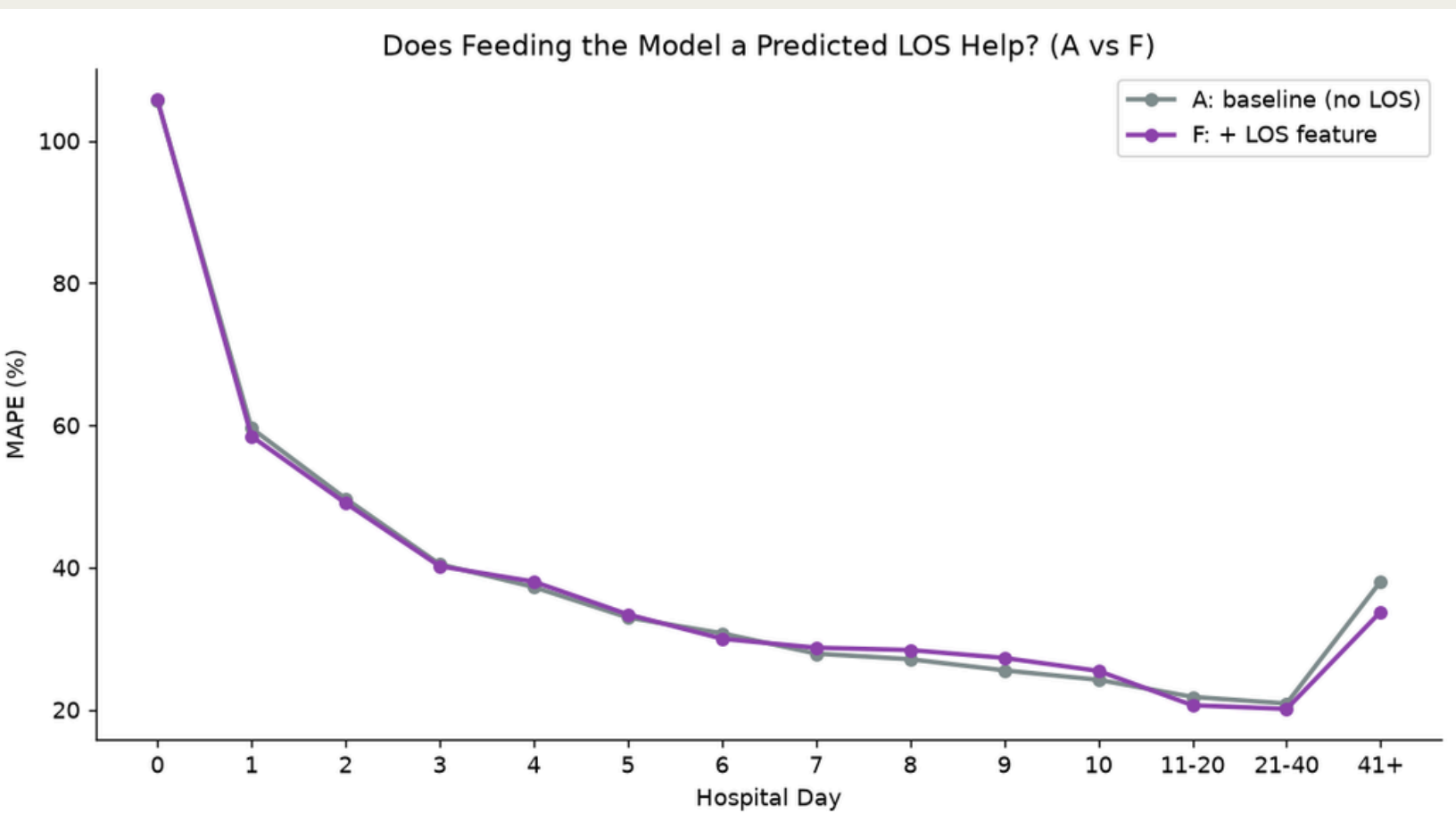


Why it rises over time here: the denominator (actual days remaining) keeps shrinking as the stay goes on — by Day 10, a patient might genuinely only have 2-3 days left. An absolute miss of even 1-2 days becomes a much bigger percentage of a small number, even though the model isn't getting worse in absolute terms. That's purely arithmetic, not the model degrading.

LOS Sub-Model — Top 25 Features (Gain vs SHAP)



Experiment 2: Does the LOS Signal Help the Cost Model?



What was tested

- **A (baseline): the original model**
- **F (+ predicted remaining stay): same model, plus the LOS sub-model's "days left" prediction fed in as an extra input**

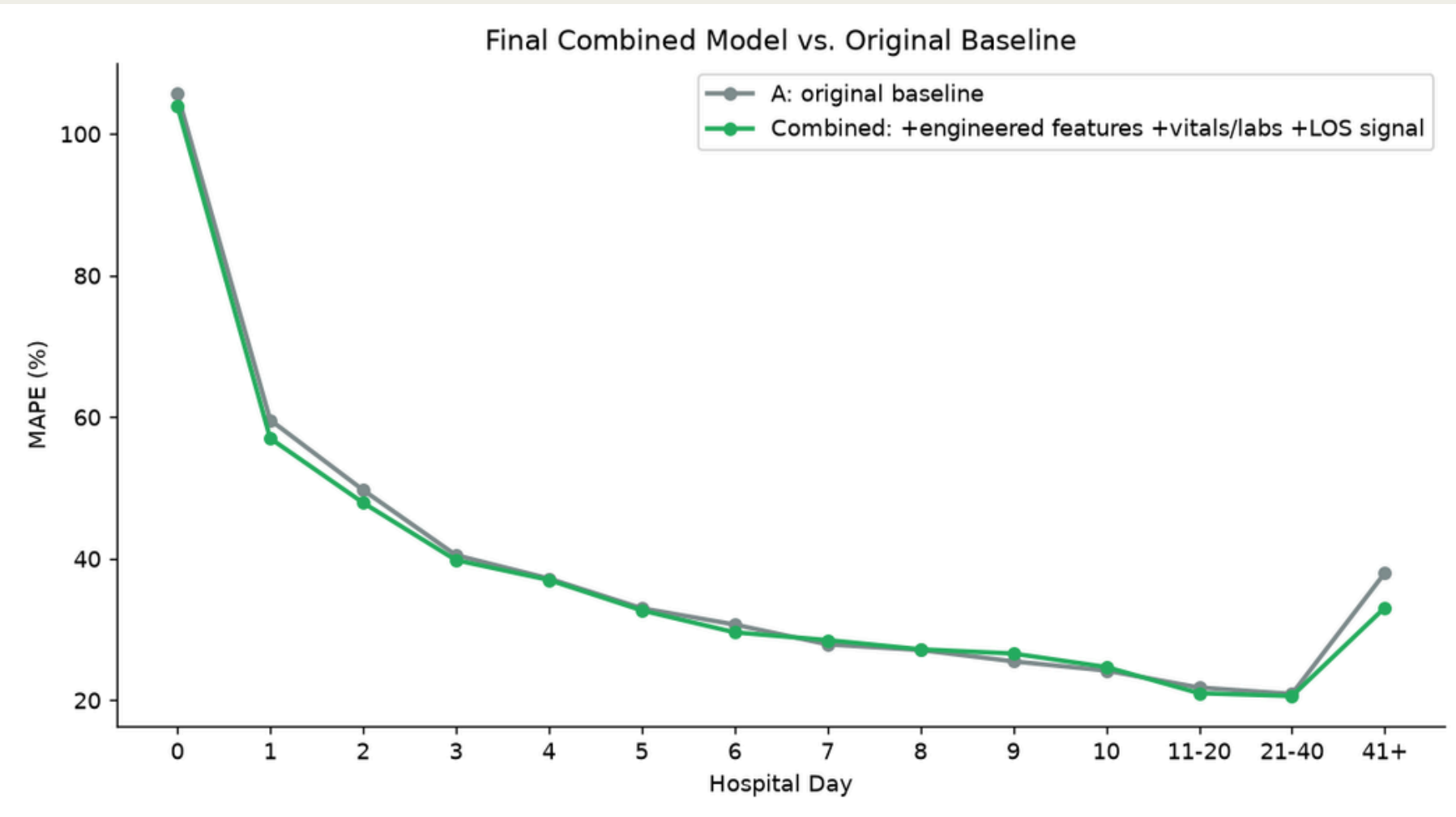
Result

- **Overall MAPE — A: 47.3% | F: 46.9%**
- **Lines are nearly identical everywhere, except Day 41+: 38.0% → 33.8%**

Conclusion

The LOS signal earns its keep only for the rarest, longest stays — not typical ones.

Combined Cost Model vs. Original Baseline



What's in the combined model

Original features + engineered rolling stats (cumulative ICU days/procedures/meds/labs, cost trend ratio) + admission vitals & labs + LOS signal — every improvement that tested positive, stacked together

Result

- Overall MAPE: 47.3% → 46.1%

XGBoost parameter tuning — did that help too?

- Tested both directions on top of this combined model: more regularization (shallower trees, stronger penalties) and more capacity (deeper trees)
- Neither meaningfully beat the current settings — best alternative was 46.0% vs. 46.1%
- Conclusion: current settings are already close to optimal — remaining gains come from features

Experiment-3: New Model – Full Test Results

TRAINING FORMULA – OLD VS. NEW

Before:

The model looked at the patient's data and directly guessed the total bill at discharge, in one shot.

Now:

Step 1 – For training only, we calculate:

remaining cost = total bill - cost accrued so far (never less than zero).

This is the answer key the model learns from.

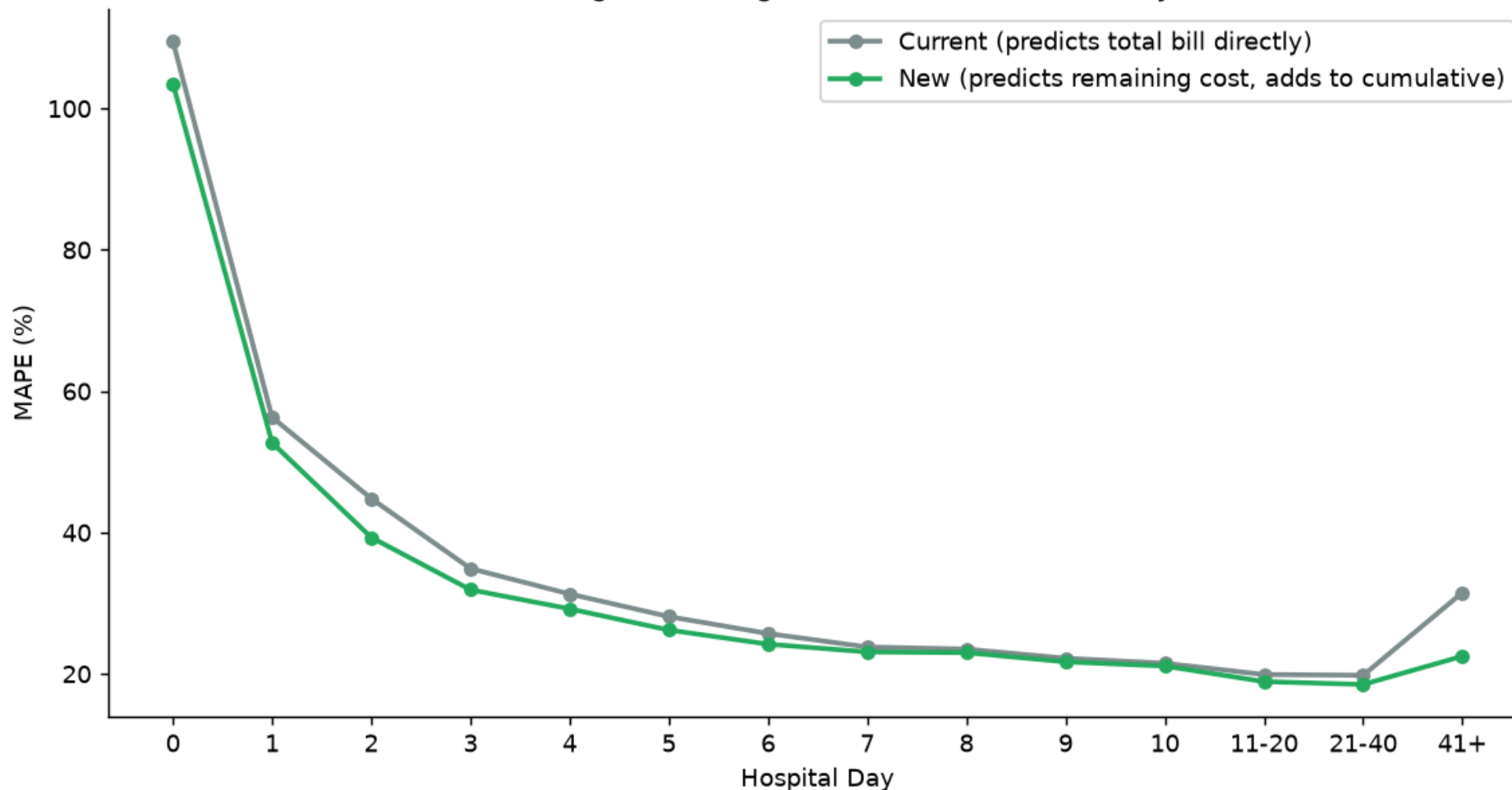
Step 2 – The model looks at the patient's data and guesses only that remaining cost – not the total.

Step 3 – We then calculate the final predicted total ourselves, in code:

Predicted total = cost accrued so far + the model's guessed remaining cost (never less than zero)

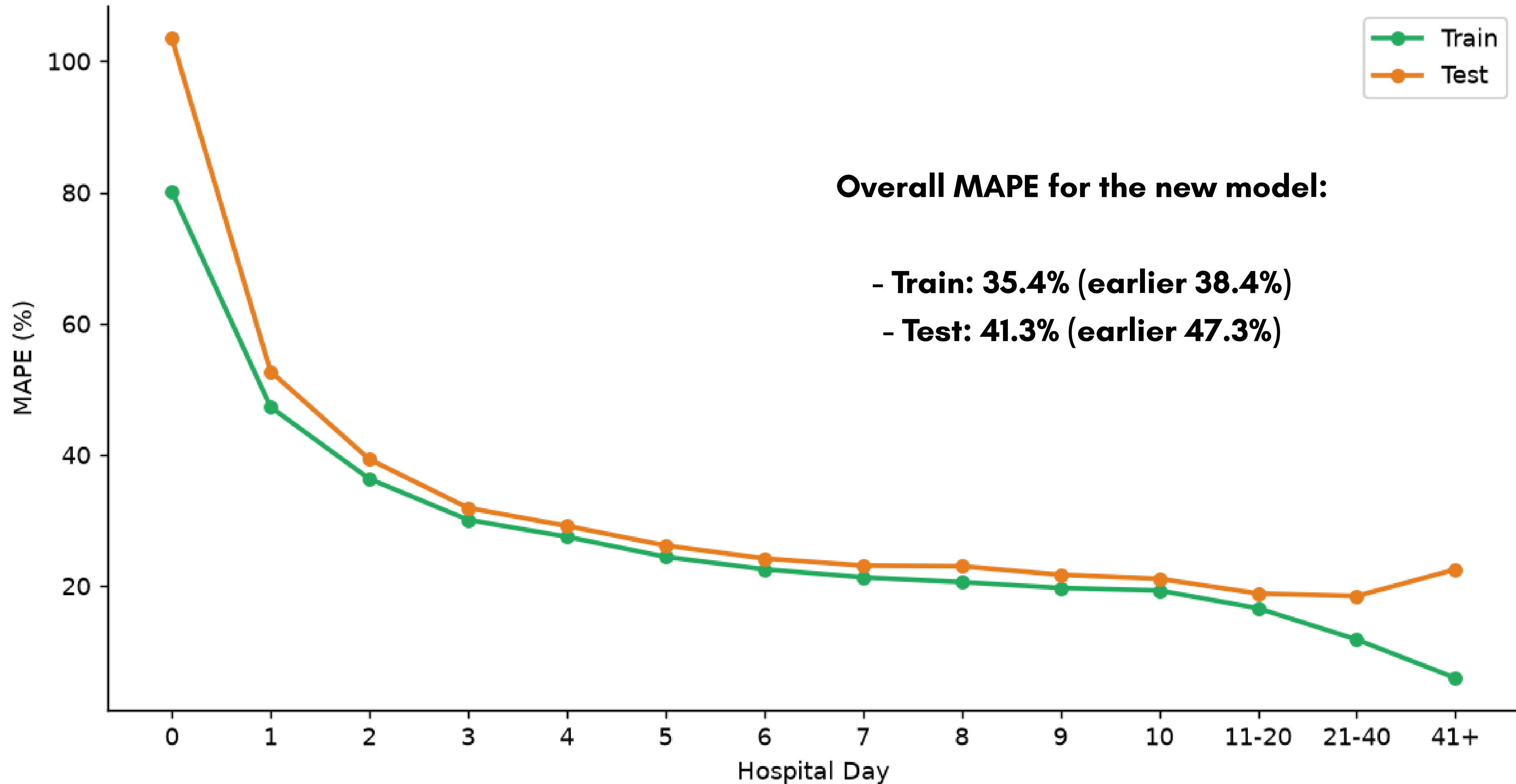
Because "cost accrued so far" is a real, already-known number, and the guessed remaining part can never go below zero, the predicted total can never fall below what's already been charged – that's guaranteed by the math itself, not a rule someone bolted on afterward.

Predicting Remaining Cost vs. Total Cost Directly



- Overall MAPE: 46.1% → 41.3% (↓ 3.2 points, ≈7% less error)
- Day 41+ MAPE: 31.5% → 22.5% (↓ 9.0 points, ≈29% less error)
- Day 0 MAPE: 109.5% → 103.5% (↓ 6.0 points)

New Model: Train vs. Test -- Overfitting Check



Moving On to the Frontend Integration